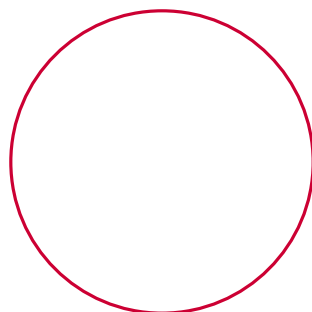


Convolutional Neural Network



Deep Learning Artificial Intelligence Report

CNN Architecture Implementation, Training Evaluation, and Reflection

1 Assignment Instructions

Purpose of the Laboratory

This laboratory introduces the foundational concepts of Convolutional Neural Networks (CNNs) through the MNIST handwritten-digit dataset. The objective is to build, train, evaluate, and reflect on a CNN while developing an understanding of how each major component contributes to image classification.

Laboratory Expectations

Important: Do not simply execute the provided code. Experiment with the model parameters and use the results to explain what the different CNN components do.

2 Reflection Questions

Assignment Direction. Answer each question in detail using the concepts demonstrated in the laboratory and observations from the experiments.

1. **Conv2D Layer.** What is the role of the Conv2D layer? How do `kernel_size` and the number of filters affect learning? *Hint: Experiment with these values in Cell 3.*
2. **MaxPooling2D Layer.** What is the purpose of MaxPooling2D? How does it contribute to model performance? *Hint: Try removing or adding a pooling layer.*
3. **One-Hot Encoding.** Why do we use one-hot encoding for the labels?
4. **Flatten Layer.** Why is the Flatten layer needed before the Dense layer?
5. **Optimizer and Loss Function.** What optimizer and loss function did you choose in Cell 4? Why is categorical cross-entropy suitable for this task? Why is Adam a good optimizer?
6. **Batch Size and Epochs.** How did you choose the batch size and number of epochs in Cell 5? What are the effects of changing these parameters? *Hint: Experiment.*
7. **Dropout.** Why is the Dropout layer included in the model?
8. **Model Architecture.** Describe the overall CNN architecture. How many convolutional layers and max pooling layers are used? What is the final Dense layer doing?
9. **Performance.** What accuracy did you achieve on the test set? Are you satisfied with the result? Why or why not? If improvement is needed, what could you try?

3 Technical Reference

Normalization	Dividing pixel values by 255 normalizes them to the range $[0, 1]$.
Reshaping	Adds a channel dimension to the images. For grayscale images, the channel dimension is 1.
One-Hot Encoding	<code>to_categorical</code> converts labels 0–9 into one-hot vectors.
Conv2D	<code>kernel_size</code> determines the filter size, such as 3×3 . The number of filters determines how many different features are learned.
MaxPooling2D	<code>pool_size</code> determines the size of the pooling window, such as 2×2 .
Optimizer	The algorithm used to update model weights during training.
Loss Function	Measures the error between model predictions and the true labels.
Batch Size	The number of samples processed during each training iteration.
Epoch	One complete pass through the entire training dataset.
Dropout	A regularization technique used to help prevent overfitting.

Run each cell, examine the output, and experiment with parameters to understand their effect on model performance.

4 Individual Self-Reflection

Michael Rivas

Personal Reflection on CNN Architecture and MNIST Classification

Model Performance

Test Accuracy: **99.23%**

Test Loss: **0.0232**

5 Individual Contributions and Reflections

Cherluk — Data Preparation

Michael Rivas — CNN Implementation

CNN Architecture Laboratory

Olu — Performance Analysis

Taki — Technical Editing and Presentation

6 Conclusion and Submission

Laboratory Completion

Congratulations on completing this notebook assignment. The laboratory builds practical experience with convolutional layers, pooling layers, activation functions, optimizers, loss functions, and training procedures while applying these concepts to handwritten-digit classification using the MNIST dataset.

Recommended Review

Submission Instructions

Deadline: February 12th